7. Visual summary (mind-map)

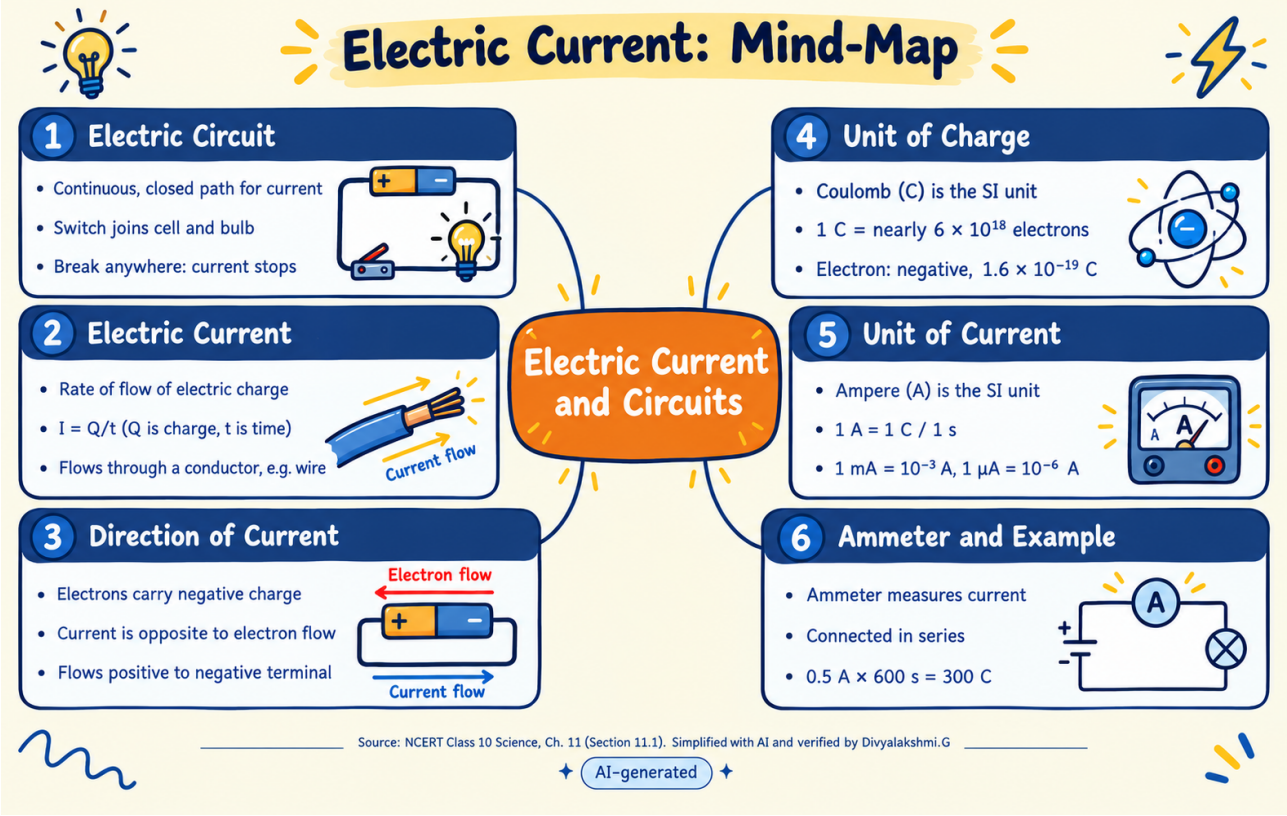
Prompt 3 was run in three tools. Outputs were compared, errors logged and a merged outline was designed as a picture.

7.1 Outline errors by tool

Tool	Main errors found
ChatGPT	"Current opposes electron flow" (should be opposite in direction); one branch had 4 points (rule: 2-3); repeated ideas in branches 1 and 2; mA and µA placed in the wrong branch; missing 6×10^{18} and 1.6×10^{-19} C.
NotebookLM	Lost the negative sign on the electron charge; formula I = Q/t placed in the direction branch; "rate of flow" omitted. Followed all structure rules.
Gemini	Dropped "nearly" before 6×10^{18}; example gave I and Q but not the time (600 s); missing 1 A = 1 C/1 s and mA/µA. Best at following structure rules, but least complete.

Process note: one Gemini paste was found to be identical to the ChatGPT outline (copied from the wrong window). It was not logged as a result, and the Gemini prompt was re-run.

7.2 Final mind-map



AI-generated image, labelled. Source: NCERT Class 10 Science, Ch. 11 (Section 11.1).

Alt text: Mind-map of electric current with six branches: electric circuit, electric current formula I = Q/t, direction of current, unit of charge, unit of current, and ammeter with a worked example of 300 coulombs, with small doodles of a circuit, wire, atom and ammeter.

7.3 Image-generation errors and fixes

Version	Problem found	Status
Version 1	Garbled "current flow" label in branch 2; no AI label and name placeholder in footer; an unlabelled resistor symbol that is not in Section 11.1; open switch drawn beside a glowing bulb	First three fixed in version 2
Version 2 (shown above)	Label readable; AI label and name added; resistor removed. The switch in branch 1 is still drawn open while the bulb glows, which contradicts "Break anywhere: current stops".	Switch still to be redrawn as closed

8. Peer feedback (status)

Status: not yet collected. A Google Form has been designed for three peers. No peer responses are included in this report, because no real responses had been received at the time of writing.

Form design

- Anonymous (no email collection); reviewers are called Peer 1, 2, 3.
- Rating questions (1-5) for the simplified text, audio and mind-map: ease of reading, clarity of key terms, match to the chapter, usefulness for revision, voice clarity, pace, numbers by ear, organisation and memorability.
- Open questions: what was confusing, which part was hard to follow, what to add or remove, any mistake or missing point.
- Overall: which format helped most; would they use it for revision.

Automation component

Form responses go to a linked Google Sheet. A Summary sheet uses formulas (AVERAGE, COUNTA, COUNTIF) to calculate average scores for each format and the number of reviewers, and a column chart compares the three formats.

Plan: choose three reviewers with different needs (one who finds long text hard, one who prefers listening, one who prefers pictures). Any real mistake they find will be added to the error log, with a note of what was changed.

9. Responsible AI and limits

- **Accuracy:** every AI output contained at least one error or lost detail. Errors ranged from broken exponents to a wrong definition of charge. Human checking against NCERT was essential.
- **Added content:** tools added facts and wording that are not in the source ("tiny", "energy is the ability to do work", a wrong source tag). This breaks the "do not add facts" rule and could mislead students.
- **Formatting damage:** LaTeX, citation numbers, emojis and tool offers leaked into outputs and would be read aloud or look broken in documents.
- **Accessibility versus completeness:** the tools that followed the short-sentence and short-point rules best were also the least complete.
- **Tone and confidence:** outputs sound confident even when slightly wrong, so a student could trust an error.
- **Image tools:** pictures can contradict the text (open switch with a glowing bulb) and can include labels that are garbled.
- **Copyright and licence:** only an open NCERT section was used, credited as the source. All AI content is labelled. The audio uses a generic AI voice, not a clone of a real person.
- **Privacy:** no personal data was uploaded. Peer feedback is anonymous.

- **Conclusion:** AI can make study material easier to access quickly, but it should support human review and never replace it.

10. Work still to do

Item	Status
Simplified text, error logs, comparison, merged final	Done
Audio script (three tools) and audio file	Done
Mind-map outlines and picture	Done except the open-switch fix in branch 1
Peer feedback: send form, collect 3 real responses, summarise	Pending
One-page how-to guide for repeating the process	Pending
Before/after analysis (original vs simplified, audio, visual) with peer scores	Pending (needs peer data)
Combined verification report and prompt library files	Pending
Live demonstration	Pending